Inverse Methods [EAS 6134]
Problem Set #4

C. Michael Haynes
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Ford 2118

Exercise 4.2 (a-f): Consider the integral eqn. & dataset from exercise 3.5:

a) discretize using simple collocation:

$$d(y) = \int_0^1 x e^{-xy} m(x)\,dx$$

let $n=20$ represent the number of segments Δx: $\Delta x = \frac{1}{n} = \frac{1}{20}$

$$\approx \sum_{j=1}^{n} \int_{x_j - \Delta x/2}^{x_j + \Delta x/2} x_j e^{-x_j y_i} \cdot m_j \, dx \qquad \text{where } m_j \equiv m(x_j)$$

$$\approx \sum_{j=1}^{n} x_j e^{-x_j y_i} m_j \, \Delta x$$

Since $\vec{d} = \underline{\underline{G}} \cdot \vec{m}$, we have that

$$G_{ij} = x_j e^{-x_j y_i} \Delta x$$

A colormap of the $\underline{\underline{G}}$ matrix is provided in the solution.

b) Since the numbers are accurate to 4 significant figures, the standard deviation will be, at most, $9.9 \cdot 10^{-5}$ from the 'true' value of the function. Then, we assume (a conservative) $\sigma = 10^{-4}$. However, we use the Theorem (4.1) provided in Aster et. al in order to establish a bound appropriate for our specific dataset) 'noise level' (i.e., error):

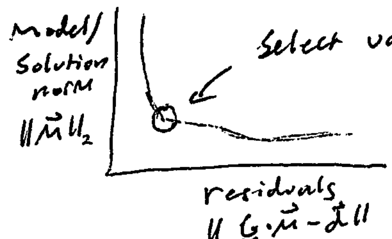
$$\delta \approx \frac{\sigma \sqrt{n}}{\|\vec{d}\|_2} \approx 5.34 \cdot 10^{-4}$$

c) – e) For parts (c–e), we perform the same process. That is we use three criteria to determine the objective function / weighting parameter α:

i) Discrepancy principle: (bound directly on residuals)

$$\| \underline{\underline{G}} \cdot \vec{m} - \vec{d} \|_2 \leq \delta \qquad (\star)$$

ii) L-curve criterion: (visually inspect / optimize tradeoff between model norm & residuals)



Select value that 'best' minimizes both for α

iii) Generalized – Cross – Validation: (GCV, see Aster sec. 4.7) the value of α that minimizes the function g,

$$g(\alpha) \equiv \frac{(m) \| \underline{\underline{G}} \cdot \vec{m}_{\alpha,L} - \vec{d} \|_2}{Tr(\underline{\underline{I}} - \underline{\underline{G}} \cdot \underline{\underline{G}}^{\#})^2}$$

Please see attached Figures. For (d), we use

$$\underline{\underline{L}}_1 \equiv \begin{pmatrix} -1 & +1 & & \\ & -1 & +1 & \\ & & -1 & +1 \\ & & & \ddots \end{pmatrix}, \text{ and for (e): } \underline{\underline{L}}_2 \equiv \begin{pmatrix} 1 & -2 & 1 & & \\ & 1 & -2 & 1 & \\ & & \ddots & \ddots & \ddots \end{pmatrix}$$

to regularize the model norm $\| \underline{\underline{L}} \cdot \vec{m} \|$. The following show our α-values for the three principles:

Tikhonov Order	α value		
	discrepancy	L-curve	GCV
0	(δ)	$1.09 \cdot 10^{-4}$	$5.43 \cdot 10^{-5}$
1	(δ)	$2.81 \cdot 10^{-4}$	$1.39 \cdot 10^{-4}$
2	(δ)	$5.69 \cdot 10^{-4}$	$2.81 \cdot 10^{-4}$

Note that for discrepancy, we use ($\star$) instead.

f) The resolution, allong with the obtained solutions via the 3 methods is included in the attached figure. The features driven by the different regularization schema are not physical (necessarily), since we are imposing conditions on the function $m(x)$, of which we know nothing about (except assuming it is continuous). The regularizers each perform the following:

T^0: Minimizes model norm $\|\vec{m}\|_2$: → seeks 'low complexity'

T^1: Minimizes model derivative norm $\|\underline{\underline{L}}_1 \cdot \vec{m}\|_2$
→ seeks 'low rate of change'

T^2: Minimizes model 2nd derivative norm $\|\underline{\underline{L}}_2 \cdot \vec{m}\|_2$
→ seeks 'low curvature'

Each of these properties are reflected in the obtained models, even though they may not be 'physical' or make sense to impose. See, for instance, how the minimized curvature solutions (T^2) exhibit a very poor fit, since $m(x)$ does curve.

The best fit is the zeroth-order Thikonov, and the symmetric resolution matrix reflects that all model parameters are equally resolved (very rough approximation) allowing the model solution to reflect the data well, similar to the truncated SVD obtained from HW3. For the higher order methods, we show the g-curve of value resolution matrix, since it is the 'best' match.